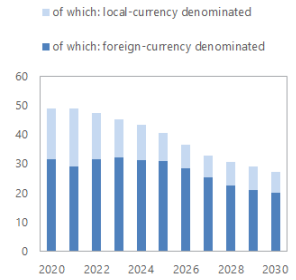
Table 2. Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–2040

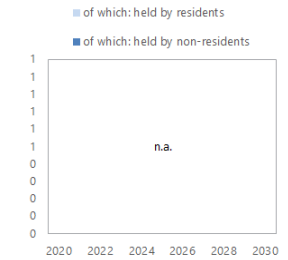
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 7/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projection
Public sector debt 1/	**50.3**	**49.1**	**44.3**	**49.1**	**49.0**	**47.5**	**45.2**	**43.6**	**40.8**	**27.3**	**12.3**	**39.3**	**39.2**
of which: external debt	**25.6**	**25.8**	**24.6**	**31.5**	**34.4**	**34.5**	**33.6**	**33.1**	**30.5**	**20.3**	**10.5**	**24.5**	**28.5**
Change in public sector debt	-1.1	-1.2	-4.8	4.8	0.0	-1.6	-2.3	-1.6	-2.8	-1.8	-1.2		
Identified debt-creating flows	**-3.1**	**-3.8**	**2.4**	**3.5**	**-1.3**	**-3.1**	**-4.4**	**-3.6**	**-4.2**	**-4.8**	**-1.9**	**0.6**	**-3.4**
Primary deficit /2	**-1.3**	**-3.0**	**-0.8**	**0.2**	**1.8**	**-0.2**	**-2.0**	**-1.5**	**-2.0**	**-3.8**	**-1.3**	**0.2**	**-1.9**
Revenue and grants	14.6	15.3	14.2	18.4	15.2	16.4	17.8	17.1	17.2	19.6	18.4	**17.8**	**17.7**
of which: grants	4.0	3.3	1.7	4.8	3.8	3.6	3.6	3.5	3.4	2.5	1.8		
Primary (noninterest) expenditure	13.3	12.3	13.4	18.7	17.0	16.2	15.8	15.6	15.2	15.8	17.0	**18.1**	**15.8**
Automatic debt dynamics	**0.0**	**-0.2**	**3.7**	**3.7**	**-2.7**	**-2.5**	**-2.0**	**-1.8**	**-1.9**	**-0.9**	**-0.5**		
Contribution from interest rate/growth differential	2.4	-1.2	3.0	2.0	-2.4	-2.1	-1.6	-1.4	-1.5	-0.7	-0.4		
of which: contribution from average real interest rate	1.2	-0.1	4.4	1.6	0.5	0.2	0.2	0.2	0.1	0.2	0.1		
of which: contribution from real GDP growth	1.3	-1.1	-1.4	0.3	-2.8	-2.3	-1.8	-1.7	-1.6	-0.8	-0.4		
Contribution from real exchange rate depreciation	-2.4	1.0	0.7	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-1.8**	**-0.6**	**-0.4**	**-0.4**	**-0.4**	**-0.4**	**-0.4**	**-0.3**	**-0.3**	**-0.2**	**-0.1**	**-0.9**	**-0.3**
Privatization receipts (negative)	-1.2	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalizatior	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	-0.5	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.3	-0.3	-0.2	-0.1		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.0**	**2.6**	**-7.2**	**3.0**	**1.0**	**1.1**	**1.7**	**1.6**	**1.0**	**2.9**	**0.6**	**1.2**	**1.7**
Sustainability indicators													
PV of public debt-to-GDP ratio 3/	**...**	**...**	**40.6**	**43.5**	**41.9**	**39.8**	**37.5**	**35.7**	**33.7**	**22.4**	**9.4**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**285.0**	**235.8**	**275.9**	**243.1**	**211.1**	**209.0**	**195.7**	**114.5**	**51.3**		
Debt service-to-revenue and grants ratio 4/	**46.5**	**18.3**	**59.1**	**56.5**	**59.7**	**55.8**	**55.5**	**64.0**	**63.3**	**52.1**	**27.7**		
Gross financing need 5/	4.1	-1.3	7.6	11.4	11.4	10.0	9.4	10.7	9.6	6.5	3.8		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-2.4	2.3	3.0	-0.8	6.1	4.9	4.0	3.8	3.8	3.0	3.2	**3.4**	**3.6**
Average nominal interest rate on external debt (in percent)	4.0	2.7	2.7	1.9	2.0	2.0	1.8	1.7	1.2	1.0	1.0	**3.2**	**1.8**
Average real interest rate on domestic debt (in percent)	2.3	-0.5	18.6	6.5	2.0	0.8	1.4	2.1	2.7	4.6	4.3	**2.9**	**3.3**
Real exchange rate depreciation (in percent, + indicates depreciati	-8.5	4.1	2.6	...	...	...	...	...	...	...	...	**4.3**	**...**
Inflation rate (GDP deflator, in percent)	-0.8	2.3	1.5	-4.7	2.1	3.4	3.5	3.3	3.2	2.9	3.2	**0.7**	**2.3**
Growth of real primary spending (deflated by GDP deflator, in perc	5.0	-5.7	12.8	37.8	-3.3	-0.1	1.3	3.0	0.8	6.1	3.2	**-1.2**	**5.6**
Primary deficit that stabilizes the debt-to-GDP ratio 6/	-0.3	-1.8	3.9	-4.5	1.8	1.4	0.3	0.2	0.8	-2.0	-0.2	**0.6**	**-0.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, central bank. Definition of external debt is Currency-based.

2/ The primary balance assumes debt relief under the CCRT as a capital grant (subject to availability of resources).

3/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

5/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

6/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

7/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

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Table 3. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–2030

(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	26.0	27.3	26.9	25.9	25.2	23.5	21.0	18.7	17.2	16.2	15.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	26.0	26.8	27.0	26.3	25.9	25.1	25.1	24.5	24.7	25.6	26.7
B. Bound Tests											
B1. Real GDP growth	26.0	**31.1**	**34.2**	**33.0**	**32.1**	29.9	26.8	23.8	21.9	20.7	19.7
B2. Primary balance	26.0	27.9	28.1	26.5	25.0	22.6	19.7	17.1	15.4	14.2	13.2
B3. Exports	26.0	**34.0**	**44.7**	**43.2**	**42.1**	**39.8**	**36.8**	**34.0**	**31.8**	29.3	27.2
B4. Other flows 3/	26.0	**30.6**	**33.0**	**31.9**	**31.0**	29.1	26.5	24.1	22.3	20.7	19.5
B6. One-time 30 percent nominal depreciation	26.0	**34.3**	29.7	28.7	27.8	25.8	22.8	19.9	18.2	17.3	16.6
B6. Combination of B1-B5	26.0	**39.3**	**41.9**	**40.5**	**39.4**	**37.1**	**33.9**	**30.8**	28.3	26.3	24.6
C. Tailored Tests											
C1. Combined contingent liabilities	26.0	26.9	25.7	23.8	22.2	19.8	17.0	14.4	12.7	11.6	10.7
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	26.0	**31.2**	**34.5**	**33.5**	**32.4**	**30.1**	27.0	24.0	21.6	19.6	18.0
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	99.2	95.4	88.0	82.5	78.8	72.6	63.4	57.1	53.3	50.8	50.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	99.2	93.7	88.4	83.6	80.9	77.7	75.6	74.9	76.4	80.2	86.5
B. Bound Tests											
B1. Real GDP growth	99.2	95.4	88.0	82.5	78.8	72.6	63.4	57.1	53.3	50.8	50.0
B2. Primary balance	99.2	97.5	91.9	84.4	78.1	69.8	59.4	52.2	47.6	44.4	42.8
B3. Exports	99.2	**166.7**	**281.1**	**264.2**	**252.7**	**237.0**	**213.5**	**200.1**	**189.1**	**176.6**	**169.5**
B4. Other flows 3/	99.2	106.8	108.0	101.3	96.9	90.1	80.0	73.6	68.9	65.0	63.1
B6. One-time 30 percent nominal depreciation	99.2	95.4	77.5	72.6	69.3	63.5	54.7	48.5	44.7	43.1	42.8
B6. Combination of B1-B5	99.2	**154.4**	112.3	**168.7**	**161.3**	**150.3**	133.8	123.5	114.6	107.9	104.5
C. Tailored Tests											
C1. Combined contingent liabilities	99.2	93.9	84.3	75.7	69.3	61.1	51.2	44.0	39.4	36.4	34.6
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	99.2	131.7	130.5	118.9	109.8	98.2	83.5	75.1	68.7	63.1	59.7
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	6.0	7.1	7.7	7.7	7.5	6.8	7.8	8.2	5.7	4.4	4.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	6.0	7.6	8.6	9.0	9.2	8.6	**10.5**	**11.6**	7.8	5.7	5.6
B. Bound Tests											
B1. Real GDP growth	6.0	7.1	7.7	7.7	7.5	6.8	7.8	8.2	5.7	4.4	4.3
B2. Primary balance	6.0	7.1	7.7	7.7	7.5	6.7	7.7	8.0	5.7	4.4	4.2
B3. Exports	6.0	**10.4**	**16.2**	**17.1**	**16.7**	**15.1**	**17.1**	**17.8**	**15.2**	**16.8**	**16.4**
B4. Other flows 3/	6.0	7.1	7.8	7.9	7.8	7.0	8.0	8.4	6.6	5.9	5.8
B6. One-time 30 percent nominal depreciation	6.0	7.1	7.7	7.5	7.4	6.6	7.7	8.1	5.6	3.5	3.5
B6. Combination of B1-B5	6.0	9.2	**12.8**	**12.8**	**12.5**	**11.3**	**12.9**	**13.5**	**12.0**	9.9	9.7
C. Tailored Tests											
C1. Combined contingent liabilities	6.0	7.1	7.6	7.6	7.4	6.6	7.6	8.0	5.5	4.2	4.1
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	6.0	8.8	9.4	9.4	9.0	7.9	8.9	9.3	7.3	6.5	6.3
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	11.5	**18.0**	**18.3**	**17.0**	**17.7**	**15.8**	**17.7**	**17.4**	12.0	8.3	7.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	11.5	**19.2**	**20.5**	**19.9**	**21.6**	**20.1**	**23.7**	**24.7**	**16.2**	11.0	10.1
B. Bound Tests											
B1. Real GDP growth	11.5	**20.5**	**23.4**	**21.7**	**22.6**	**20.1**	**22.5**	**22.2**	**15.3**	10.6	9.8
B2. Primary balance	11.5	**17.9**	**18.3**	**17.1**	**17.7**	**15.7**	**17.5**	**17.0**	11.9	8.5	7.6
B3. Exports	11.5	**18.6**	**20.2**	**19.8**	**20.5**	**18.3**	**20.1**	**19.8**	**16.6**	**16.7**	**15.4**
B4. Other flows 3/	11.5	**18.0**	**18.7**	**17.6**	**18.3**	**16.3**	**18.2**	**17.9**	13.9	11.3	10.4
B6. One-time 30 percent nominal depreciation	11.5	**22.6**	**23.0**	**20.9**	**21.8**	**19.4**	**21.8**	**21.6**	**14.8**	8.5	7.9
B6. Combination of B1-B5	11.5	**20.5**	**23.4**	**21.7**	**22.5**	**20.1**	**22.3**	**22.0**	**19.2**	**14.5**	13.4
C. Tailored Tests											
C1. Combined contingent liabilities	11.5	**17.9**	**18.2**	**16.8**	**17.4**	**15.4**	**17.3**	**17.0**	11.6	8.0	7.4
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	11.5	**23.9**	**24.5**	**22.9**	**22.9**	**19.5**	**20.5**	**19.3**	**14.9**	12.2	11.0
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

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Table 4. Chad: Sensitivity Analysis for Key Indicators of Public Debt, 2020–2030

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	**43.5**	**41.9**	**39.8**	**37.5**	**35.7**	33.7	30.3	27.3	25.2	23.7	22.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	**43**	**43**	**43**	**44**	**45**	**46**	**46**	**46**	**47**	**49**	**51**
B. Bound Tests											
B1. Real GDP growth	**43**	**84**	**93**	**92**	**91**	**90**	**88**	**86**	**85**	**85**	**85**
B2. Primary balance	**43**	**45**	**45**	**42**	**41**	**39**	**35**	32	30	28	26
B3. Exports	**43**	**47**	**55**	**52**	**50**	**48**	**44**	**41**	**38**	35	33
B4. Other flows 3/	**43**	**45**	**46**	**43**	**42**	**39**	**36**	33	30	28	26
B6. One-time 30 percent nominal depreciation	**43**	**87**	**83**	**79**	**76**	**72**	**67**	**63**	**59**	**56**	**54**
B6. Combination of B1-B5	**43**	**44**	**45**	**43**	**43**	**42**	**39**	**36**	34	33	31
C. Tailored Tests											
C1. Combined contingent liabilities	**43**	**57**	**54**	**52**	**49**	**47**	**42**	**39**	**36**	34	32
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	**43**	**77**	**81**	**85**	**88**	**89**	**87**	**86**	**85**	**85**	**85**
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	235.8	275.9	243.1	211.1	209.0	195.7	169.2	147.4	140.8	123.9	114.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	236	280	259	244	257	258	247	240	252	247	251
B. Bound Tests											
B1. Real GDP growth	236	532	532	489	506	498	465	441	455	427	420
B2. Primary balance	236	295	272	238	237	224	196	172	166	146	135
B3. Exports	236	312	335	293	292	276	245	219	211	183	166
B4. Other flows 3/	236	297	280	245	243	229	200	176	169	148	135
B6. One-time 30 percent nominal depreciation	236	587	514	453	451	426	380	343	334	298	279
B6. Combination of B1-B5	236	291	270	240	248	240	216	193	189	169	158
C. Tailored Tests											
C1. Combined contingent liabilities	236	378	332	290	288	270	236	209	200	176	162
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	236	609	579	550	568	547	496	453	466	436	428
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	56.5	59.7	55.8	55.5	64.0	63.3	67.4	66.1	62.4	54.7	52.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	56	61	53	56	71	72	82	87	87	78	78
B. Bound Tests											
B1. Real GDP growth	56	65	68	70	81	82	88	89	87	78	76
B2. Primary balance	56	59	59	59	61	59	64	64	61	54	51
B3. Exports	56	60	56	57	65	64	68	67	65	61	58
B4. Other flows 3/	56	60	56	56	65	64	68	67	64	57	54
B6. One-time 30 percent nominal depreciation	56	58	58	54	65	64	68	67	61	52	49
B6. Combination of B1-B5	56	60	58	58	67	66	70	70	66	58	56
C. Tailored Tests											
C1. Combined contingent liabilities	56	59	54	52	59	58	63	64	60	53	51
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	56	73	69	70	90	89	90	85	82	75	73
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

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